SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	G. Poojitha	**Reg. No.:**	192425337
Section / Batch:	Slot A / 2024	**Date:**	5-10-2026.

Code of Conduct

I G. Poojitha (192425337) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Poojitha.

QUESTIONS: 5

Total Marks:50

Annexure A

Short Answer Test

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position. **Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.**

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.

Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.

Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.

Analyze how the fundamental stages of a Computer Vision system can be used for this application.

Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.

Create a suitable Computer Vision pipeline for this application using image acquisition, preprocessing, segmentation, feature extraction, and classification. Justify the purpose of each stage.

Smart Traffic Monitoring System

A smart Traffic Monitoring System uses Computer vision to monitor vehicles, pedestrians and traffic conditions through cameras.

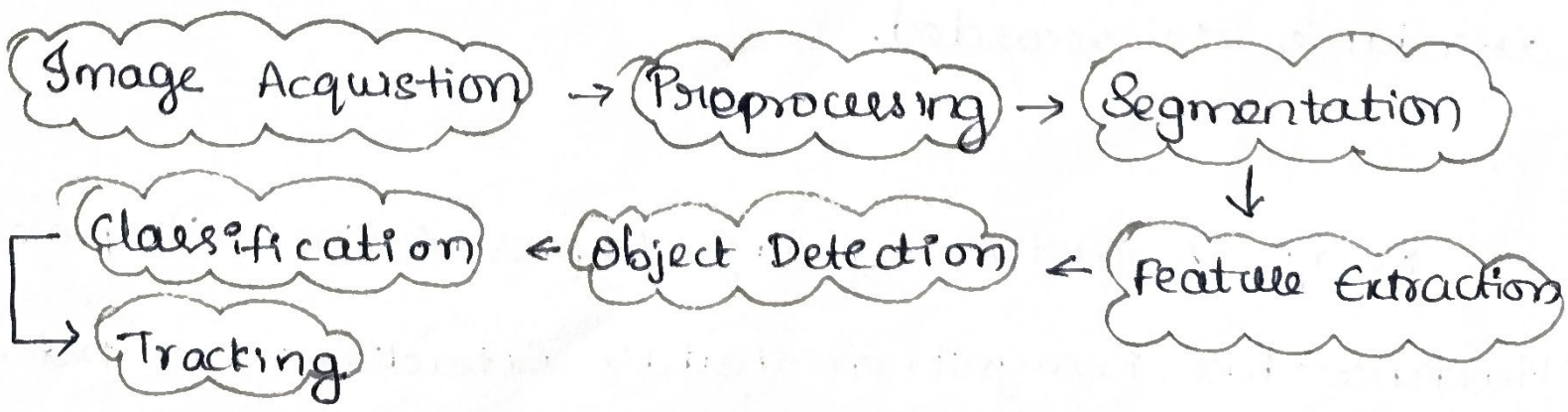
The first stage is image acquistion, where cameras capture images or video of traffic junctions. The images may have different resdutions, lighting conditions and camera angles.

Next, preprocessing is performed to improve image quality. Noise removal, ~~resizing~~, brightness correction, constrast enhancement and normalization can be used.

In segmentation, the image is divided into important regions such as roads, vehicles and pedestrians. Feature Extraction then identifies properties such as shape, size, color, edges and texture.

The system performs object detection to locate vehicles and pedestrians using bounding boxes. classification identifies the detected objects as cars, buses, trucks, motorcycles, bicycles or pedestrians.

Pipeline:



2) Automated Attendance System

An automated attendance system uses cameras to identify students through their faces and automatically record attendance.

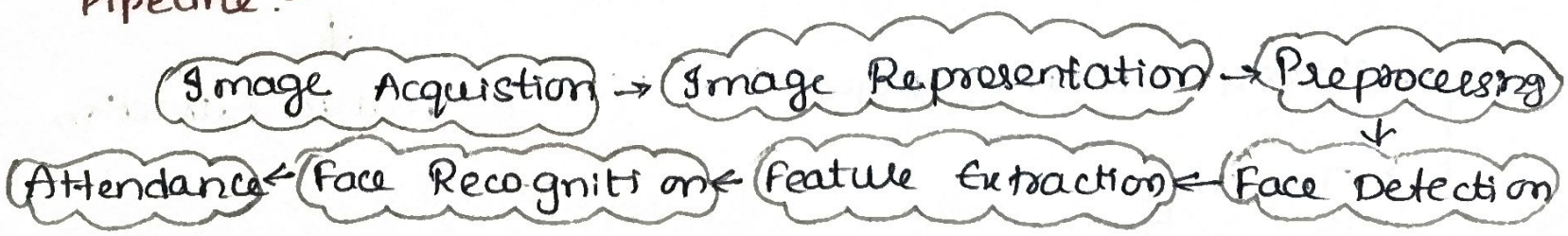
First, cameras perform image acquistion by capturing classfoom images containing multiple students. The image is represented using pixels and may contain RGB or grayscale information.

During preprocessing, techniques such as grayscale conversion, resizing, noise removal, constrast enhancement, and brightness correction are applied to improve image quality.

Next, face detection identifies the faces present in the image. Feature extraction obtains important facial characteristics such as eyes, nose, mouth and overall face structure. These features are converted into a suitable representation for recognition.

In face recognition, the extracted features are compared with registered student data. If a match is found, the student's identity is determined and attendance is recorded.

Pipeline :-



Medical Image Analysis System.

A Medical Image Analysis System helps doctors analyze X-ray images and identify abnormal regions.

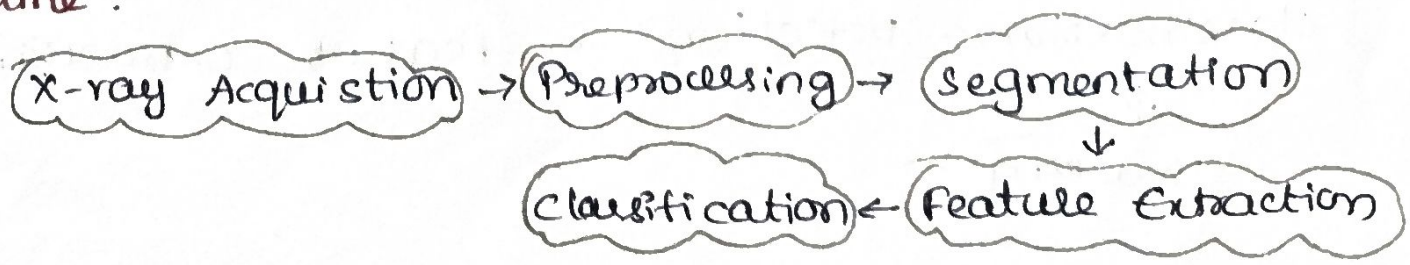
Image resolution determines the amount of detail available. Higher resolution can help identify small abnormalities. Pixel intensity represents brightness and helps distinguish different tissues and structures.

X-ray images may contain noise and poor contrast, so preprocessing is required. Noise filtering, constrast enhancement, brightness correction and normalization improve image quality.

In segmentation, the important region such as a bone, lung or suspected abnormal area is separated from the background.

After segmentation, feature extraction identifies properties such as shape, size, texture, edges and intensity. These features are then used for classification, where the system can classify regions as normal or abnormal.

Pipeline :



4) Security Surveillance System

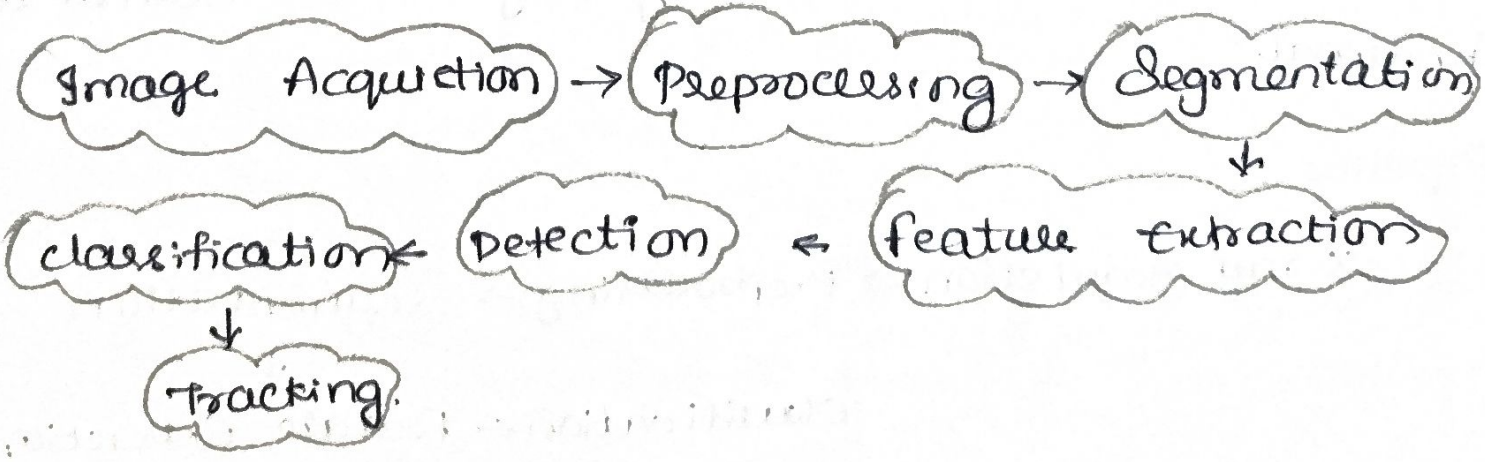
A security surveillance system uses cameras and computer vision to identify people and objects in shopping malls, corridors, entrances and parking areas.

The first stage is image acquisition, where cameras capture images or video. Preprocessing improves these images using noise removal, resizing, brightness correction and constrast enhacement.

Next, segmentation separates important objects such as people and vehicles from the background. Feature extraction identifies properties such as shape, colour, texture and edges.

Object detection and classification are then used to locate and identify people, vehicles, bags and other objects. Tracking can follow a person's movement across consecutive frames.

Pipeline:



Smart Agriculture Monitoring

A smart Agriculture Monitoring system uses drone cameras and computer vision to identify healthy plants, damaged plants, weeds and diseased areas.

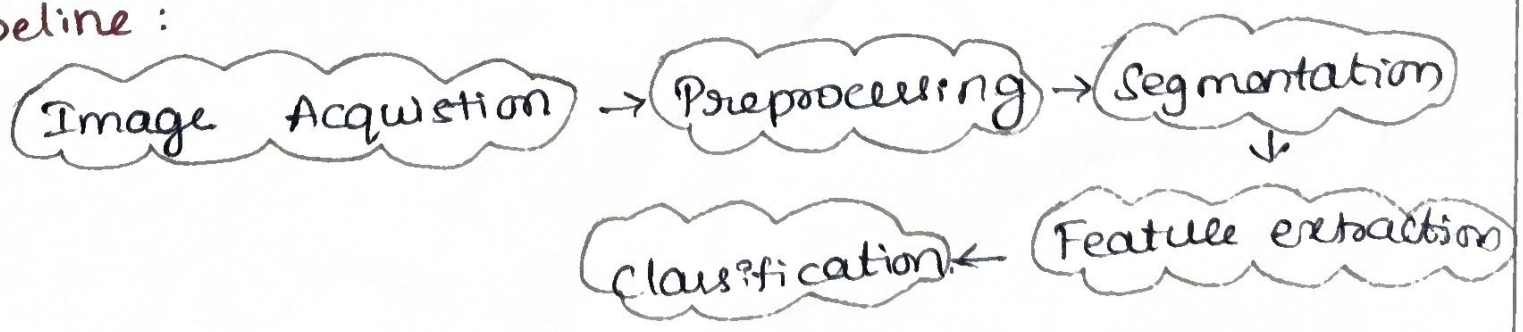
First, Image acquistion is performed using drone cameras to capture large agricultural fields. Image may contain shadows, different lighting and plants of different sizes.

During preprocessing, noise is removed and brightness, constrast and image size are adjusted to improve quality.

Next, segmentation separates plants, soil, weeds, and affected regions from the background. Feature extraction identifies features such as colour, shape, texture, size and leaf patterns.

Finally, classification identifies the regions as healthy plants, damaged plants, weeds or disease-affected plants.

Pipeline :



Rubrics for Calculation

Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date: 05/10/26	Date:	Date: